9 Data and Code Availability

The data (weekly ETF returns) and the full code are available at `https://github.com/ezellabra/ml4e-connectedness`. The pipeline is written in Python; NETS and FNETS are estimated in R (packages `nets` 0.9.1 and `fnets` 0.1.6) through a command-line bridge. All parameters are in `config.yaml` and the whole analysis is reproduced with `make all` (or `python run_all.py`).

A Robustness

Table 3. Robustness: mean total connectedness (%) by method, share of DY connectedness lost once the common factor is removed (FNETS), and FNETS common-variance share.

Scenario	Windows	DY	Lasso VAR	NETS	FNETS (idio.)	Lost w/o factor (%)	Common var. (%)
Baseline (w=104, H=4)	179	81.0	80.0	61.5	47.2	41	58
H=2	179	80.9	80.0	61.5	47.2	41	58
H=8	179	81.0	80.1	61.5	47.2	41	58
w=78	185	81.2	79.6	57.8	50.5	38	58
w=130	172	81.1	80.5	63.4	46.2	43	58
Excluding SPY	179	80.0	78.9	59.9	45.9	42	58
Excluding windows touching 2020	141	79.7	78.6	58.8	47.2	41	55

Table 4. Static connectedness: one estimate on the whole period (full sample) and on the subsamples before and after 2020. These are single long-sample estimates, not averages over rolling windows, so they differ from Tables 2 and 3.

Sample	Method	Total conn. (%)	LR p.corr. dens. (%)
Full sample	DY	81.9	100.0
Full sample	Lasso VAR	82.1	100.0
Full sample	NETS	73.5	60.1
Full sample	FNETS (idio.)	34.2	0.0
2011-2019	DY	83.0	100.0
2011-2019	Lasso VAR	82.8	100.0
2011-2019	NETS	69.6	60.1
2011-2019	FNETS (idio.)	34.1	2.0
2021-2026	DY	73.5	100.0
2021-2026	Lasso VAR	73.7	100.0
2021-2026	NETS	56.0	52.3
2021-2026	FNETS (idio.)	39.6	13.7

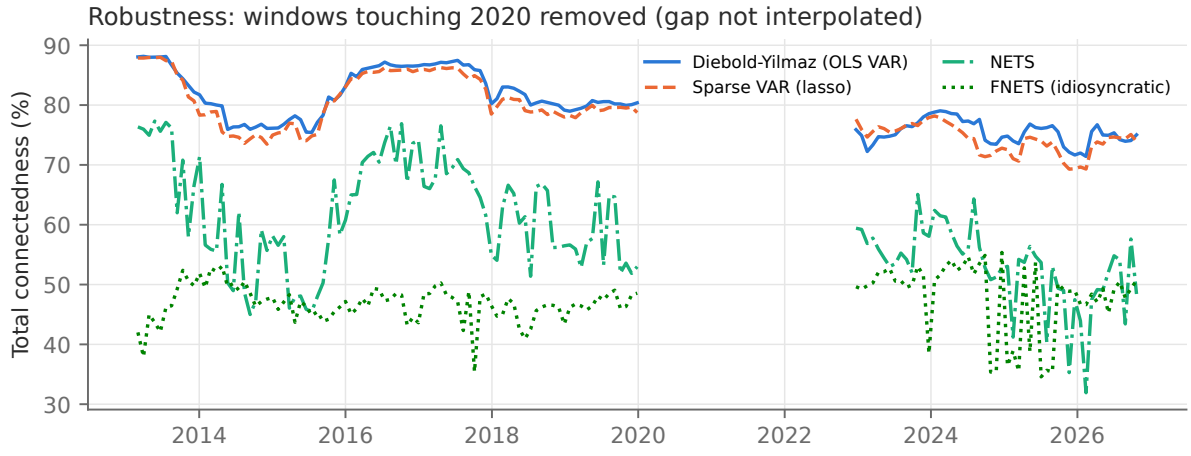


Fig. 6. Total connectedness excluding every window that contains a week of 2020.

Table 3 reruns every method under alternative choices. The levels of Diebold–Yilmaz and lasso connectedness hardly move: between 79.7% and 81.2% and between 78.6% and 80.5%